

Library Management System — Output Screenshots

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Output Q1 — Multi-table Join (Currently Issued Books)

```
mysql> SELECT m.name AS member_name, b.title AS book_title,
-> l.issue_date, l.due_date
-> FROM Loans l
-> JOIN Members m ON m.member_id = l.member_id
-> JOIN Books b ON b.book_id = l.book_id
-> WHERE l.return_date IS NULL
-> ORDER BY l.due_date;
```

member_name	book_title	issue_date	due_date
Priya Sharma	Introduction to Algorithms	2026-08-20	2026-09-03
Rohan Das	The Great Gatsby	2026-08-25	2026-09-08
Sneha Iyer	A Brief History of Time	2026-08-27	2026-09-10
Karan Mehta	Clean Code	2026-08-29	2026-09-12

4 rows in set (0.00 sec)

Output Q2 — View: vw_overdue_loans

```
mysql> CREATE VIEW vw_overdue_loans AS
-> SELECT m.name AS member_name, b.title AS book_title,
-> DATEDIFF(CURDATE(), l.due_date) AS days_overdue
-> FROM Loans l
-> JOIN Members m ON m.member_id = l.member_id
-> JOIN Books b ON b.book_id = l.book_id
-> WHERE l.due_date < CURDATE() AND l.return_date IS NULL;
```

Query OK, 0 rows affected (0.02 sec)

```
mysql> SELECT * FROM vw_overdue_loans;
```

member_name	book_title	days_overdue
Priya Sharma	Introduction to Algorithms	1

1 row in set (0.00 sec)

Output Q3 — Function: fn_compute_fine

```
mysql> DELIMITER //
mysql> CREATE FUNCTION fn_compute_fine(p_due_date DATE, p_return_date DATE)
    -> RETURNS DECIMAL(10,2) DETERMINISTIC
    -> BEGIN
    -> DECLARE v_days INT;
    -> DECLARE v_fine DECIMAL(10,2);
    -> IF p_return_date IS NULL OR p_return_date <= p_due_date THEN
    -> RETURN 0.00;
    -> END IF;
    -> SET v_days = DATEDIFF(p_return_date, p_due_date);
    -> SET v_fine = v_days * 2.00;
    -> RETURN v_fine;
    -> END //
```

Query OK, 0 rows affected (0.01 sec)

```
mysql> DELIMITER ;
mysql> SELECT fn_compute_fine('2026-09-03','2026-09-08') AS fine_amount;
+-----+
| fine_amount |
+-----+
| 10.00       |
+-----+
1 row in set (0.00 sec)
```

Output Q4 — Trigger: trg_check_available_copies

```
mysql> DELIMITER //
mysql> CREATE TRIGGER trg_check_available_copies
    -> BEFORE INSERT ON Loans
    -> FOR EACH ROW
    -> BEGIN
    -> DECLARE v_copies INT;
    -> SELECT available_copies INTO v_copies
    -> FROM Books WHERE book_id = NEW.book_id;
    -> IF v_copies <= 0 THEN
    -> SIGNAL SQLSTATE '45000'
    -> SET MESSAGE_TEXT = 'No available copies for this book';
    -> END IF;
    -> END //
```

Query OK, 0 rows affected (0.01 sec)

```
mysql> DELIMITER ;
mysql> SELECT book_id, title, available_copies FROM Books WHERE book_id = 7;
+-----+-----+-----+
| book_id | title                | available_copies |
+-----+-----+-----+
| 7       | Data Structures in C | 0                |
+-----+-----+-----+
1 row in set (0.00 sec)

mysql> INSERT INTO Loans (book_id, member_id, issue_date, due_date)
    -> VALUES (7, 3, '2026-09-03', '2026-09-17');
ERROR 1644 (45000): No available copies for this book
```